

## The dynamic sector

Kinetic energy as a  $\xi$ -fractal fraction of the static ( $\tilde{T} \cdot m = 1$ ): problem statement, narrowing, and the structural lead

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## Contents

|     |                                                                                 |   |
|-----|---------------------------------------------------------------------------------|---|
| .1  | Where the question comes from . . . . .                                         | 2 |
| .2  | The fixed foundation: the reversible Hilbert-space basis . . . . .              | 2 |
| .3  | What is fixed . . . . .                                                         | 3 |
| .4  | The single data point: the neutrino . . . . .                                   | 3 |
| .5  | Narrowing the $\xi$ -power: $p = 2$ . . . . .                                   | 3 |
| .6  | The structural lead: $m_{\text{window}} = \xi \cdot \omega$ . . . . .           | 4 |
| .7  | What remains to be derived . . . . .                                            | 4 |
| .8  | Closing the scale question: the lab anchors select $E_0$ . . . . .              | 4 |
| .9  | The $\xi^1$ mechanism: lifting the zero mode by the dimension deficit . . . . . | 5 |
| .10 | The signal-engineering reading: AM sidebands and mode-locking . . . . .         | 5 |
| .11 | Recursion as feedback: loop gain, regimes, discipline . . . . .                 | 7 |
| .12 | Relation to the static sector . . . . .                                         | 8 |
| .13 | Reproducibility . . . . .                                                       | 9 |

## Abstract

Through the duality  $\tilde{T} \cdot m = 1$ , FFGFT separates two energy types: *static* (rest mass = topologically frozen winding, the crystallographic  $Z_3$  core) and *kinetic* (free; for massless states the only form,  $E = hf$ ). The static sector is derived and computed ( $Z_3$  mass operator + mass circle; Doc. 285, BD12-XCORE). The dynamic sector is *not*: the derived rule that computes the kinetic part, via the duality, as a  $\xi$ -fractal fraction of the static one is missing. This document is a *problem-statement document* – it records what is fixed (the duality places the window in the kinetic sector), the single existing data point (neutrino,  $m_\nu \sim \xi^2/2 m_e$ , as an ansatz), the narrowing of the  $\xi$ -power to  $p = 2$ , and the structural lead  $m_{\text{window}} = \xi \cdot \omega$ . What remains to be derived is marked explicitly as open. Prefixed is the fixed, reversible Hilbert-space foundation (Doc. 282/270/230) the sector sits on; its actual aim is to find, out of the Hilbert space, the complementary description that maps via  $D_f = 3 - \xi$  and  $\xi$  onto SI units.

## 1 Where the question comes from

In Doc. 255,  $\tilde{T} \cdot m = 1$  enforces two energy types. The rest mass is *static*: a topologically stable winding number on  $T^4$ , fixed in the compact picture by the  $Z_3$  mass operator ( $r = \sqrt{2}$ ,  $\theta = 2/9$ ). The kinetic energy is *free*; for a massless particle it is the only form ( $E = hf$ ). "Mass is frozen kinetic energy."

Doc. 285 shows that the empty recursion windows ( $3'$ ,  $3''$ , ...) carry no *new* masses but structure; and that a *dynamics* filling a window supplies the *kinetic* type – a filled window is photon-/neutrino-like, not a massive triple. There the gap was named: the *derived* rule computing the kinetic part, via the duality, as a  $\xi$ -fractal fraction of the static one is missing. This document takes up exactly that gap. It is the "further work" on the dynamic sector announced in Doc. 285.

## 2 The fixed foundation: the reversible Hilbert-space basis

Before developing the dynamic sector, let the fixed foundation it sits on be recorded. The FFGFT QM translation (Doc. 282) is closed in three pieces:

1. **The unambiguous Hilbert map** (Type III, Doc. 270):  $\mathcal{H} = L^2(T^4) \otimes \mathbb{C}^3$ , states = fiber-valued fields, QM operation = unitary parallel transport, entanglement = holonomy around cycles (Doc. 230), three masses  $\leftrightarrow (M, r, \theta)$ . A *bijection* – fully reversible.
2. **The time-free mass form**: mass sits on a separate tensor factor; CHSH =  $2\sqrt{2}$  stays identical with and without the  $\mathbb{C}^3$  fiber to  $4 \cdot 10^{-16}$  – no static QM result changes. Time is *not computed* here; the reduced dynamics is time-local, unitary, Markovian.
3. **The time form**: adds exactly *one* effect – non-Markovian memory in the reduced dynamics (backflow  $0 \rightarrow 0.53$ , CP-divisibility violated) – which itself still lies within QM: a traced-out unitary bath reproduces it (Doc. 282/283). At the full level, therefore, unitary/reversible.

Reversibility is precise *per type* (Doc. 270): Type III (Hilbert) fully reversible; Type I (mass circle  $\rightarrow$  time, "time taken into the space") *almost* reversible – lossless in mode content, only the winding number stays ambiguous; Type II (spatial projection  $D3 \rightarrow D0$ ) *not* reversible. The only genuine irreversibility is the spatial projection – the measurement / decompactification (Doc. 270/284) –, not the time-inclusion. The non-Markovian memory is not a break of reversibility but the reduced view.

**Aim of the dynamic sector.** On this fixed, reversible foundation, the actual task is the *Hilbert-space extension and its translation: another description out of the Hilbert space* – the complementary, decompactified/projected side that maps computationally, via  $D_f = 3 - \xi$  and  $\xi$ , onto SI units. The SI map is exactly the measurement/projection side (Type I/II), and  $D_f$  and  $\xi$  are the bridge quantities (Doc. 270/249/133). The dynamic sector is thus not a departure from the foundation but its unrolling into measurable units – and the sought kinetic  $\xi$ -fraction is one building block of exactly that unrolling.

### .3 What is fixed

The duality fixes the *placement*, not the size. With  $\tilde{T} = 1/\max(m, \omega)$  (Doc. 067, 080), an almost-massless window state has  $\omega \gg m$ , hence  $\tilde{T} = 1/\omega$ : the state sits in the *kinetic* sector. That is secure. So is the conceptual separation:

|        | static               | kinetic                      |
|--------|----------------------|------------------------------|
| form   | $E_0 = mc^2$ (rest)  | $E = hf$ (free)              |
| seat   | $Z_3$ core (winding) | window, when a dynamics runs |
| time   | $\tilde{T} = 1/m$    | $\tilde{T} = 1/\omega$       |
| status | derived (Doc. 285)   | <i>open</i> (this document)  |

A distinction between two kinetic quantities matters: the ordinary kinetic energy of the electron,  $(\gamma - 1)mc^2$  (Doc. 080), is *not*  $\xi$ -suppressed. The  $\xi$ -fractal fraction sought here is the *window rest-mass* of an otherwise massless state – not the kinetic energy of a massive one.

### .4 The single data point: the neutrino

Doc. 047 treats the neutrino as a “near-massless photon”:  $E^2 = (pc)^2 + (\sqrt{\xi^2/2} mc^2)^2$ , with  $v_\nu = c(1 - \xi^2/2) \approx 0.9999999911c$  and  $m_\nu \sim \frac{\xi^2}{2} m_e$ . This is the only existing value for a window rest-mass as a  $\xi$ -power of a core mass. It is, however, explicitly an *ansatz* (from the velocity/photon analogy), not *derived* from the duality. Doc. 204 additionally supplies a  $\xi$ -correction to the kinetic energy ( $+0.69\xi_0$ ). There are no further data points; in particular, a single value is not enough for the  $\Delta m^2$  spectrum.

### .5 Narrowing the $\xi$ -power: $p = 2$

With the ansatz  $m_{\text{window}} = C \cdot \xi^p \cdot m_{\text{ref}}$ , tested against the bounds (reproducing the  $\xi^2/2$  value; the photon limit; dimensional consistency; an oscillation floor  $\gtrsim \sqrt{\Delta m_{\text{atm}}^2} \approx 49$  meV; a direct kinematic bound  $m_\nu < 0.45$  eV (KATRIN 2025, 90 % CL, model-independent)), only  $p = 2$  is compatible with the meV scale:  $p = 1$  gives  $\mathcal{O}(10\text{--}70)$  eV (far above the bound),  $p \geq 3$  falls below every anchor. So  $p = 2$  is the *unique* order-of-magnitude compatible power – a genuine order-of-magnitude bound, not a fit.

**Numerology warning.** Within  $p = 2$ , the coefficient ( $\frac{1}{2}$ ) and the reference mass are *not* fixed by the single anchor. Different references land somewhere in the neutrino window (e.g.  $\xi^2/2 \cdot m_e \approx 4.5$  meV,  $\xi^2 \cdot m_e \approx 9$  meV,  $\xi^2/2 \cdot \sqrt{m_e m_\mu} \approx 65$  meV). This is *not* evidence but the P35 trap: coefficient and reference need a derivation, not tuning. Computed: `kinetik_xi_anteil.py`.

## .6 The structural lead: $m_{\text{window}} = \xi \cdot \omega$

The Doc. 047 velocity expression  $v_v = c(1 - \xi^2/2)$  is exactly equivalent to a  $\xi^1$  relation. For  $m \ll p$  one has  $v/c = 1 - \frac{1}{2}(m/p)^2$ ; substituting gives  $(m/p)^2 = \xi^2$ , hence

$$m_{\text{window}} = \xi \cdot p \quad (\approx \xi \cdot \omega \text{ via } \tilde{T} \cdot m = 1). \quad (1)$$

The " $\xi^2$ " in the notation "mass =  $\xi^2/2 \cdot m_e$ " is thus presumably the *square image* of a fundamental  $\xi^1$  relation: the static rest-mass is  $\xi$  times the kinetic scale. This is the natural anchor for a derivation *via the duality*, because  $\omega = 1/\tilde{T}$  is exactly the duality quantity. The open question therefore shifts from "why  $\xi^2$ ?" to "why  $\xi^1$  between rest-mass and kinetic scale, and what sets  $\omega$  absolutely?".

## .7 What remains to be derived

Three points are explicitly open:

1. **The reference/scale – largely closed** (next section). The two lab anchors select the reference as the *single* FFGFT scale  $E_0 = \sqrt{m_e m_\mu}$ ; in its units  $\omega/E_0 = \xi/2$  and  $m_{\text{window}}/E_0 = \xi^2/2$ . Residual: the *one* structural statement  $\omega = \xi E_0$  (exactly one fractal  $\xi$ -power).
2. **The coefficient.** The factor  $\frac{1}{2}$  in  $\xi^2/2$  is not derived; possibly it is purely the velocity square  $(\frac{1}{2}(m/p)^2)$ , i.e. no free factor at all.
3. **The spectrum.** A single  $\xi$ -fractal value sets the *scale*, not the *spectrum* ( $\Delta m_{\text{sol}}^2$ ,  $\Delta m_{\text{atm}}^2$ , mixing angles). That demands the specified  $T^7$  dynamics (the second open computation, Doc. 285).

## .8 Closing the scale question: the lab anchors select $E_0$

The first open point – what sets  $\omega$  absolutely – closes further than a free parameter would. In FFGFT there is only *one* absolute scale; everything else is dimensionless ratios. The question is therefore not "which number" but "which of the existing scales". With  $m_{\text{window}} = \frac{\xi^2}{2} \cdot \text{reference}$ , the two *lab* anchors (oscillation floor  $\gtrsim 49$  meV; KATRIN  $< 0.45$  eV) bracket the reference:

| reference                | $m_{\text{window}}$ | floor $\gtrsim 49$ meV   | KATRIN $< 0.45$ eV       |
|--------------------------|---------------------|--------------------------|--------------------------|
| $m_e$                    | 4.5 meV             | <i>fails</i> (too light) | yes                      |
| $E_0 = \sqrt{m_e m_\mu}$ | 65 meV              | yes                      | yes                      |
| $M \approx 314$ MeV      | 2.8 eV              | yes                      | <i>fails</i>             |
| $m_\mu$                  | 940 meV             | yes                      | <i>fails</i> (too heavy) |

$m_e$  is too light,  $M$  and  $m_\mu$  too heavy – *only*  $E_0 = \sqrt{m_e m_\mu}$  fits, and that is exactly the one natural FFGFT scale (Doc. 282,  $\lambda_e \lambda_\mu = E_0$ ). The reference is thus not a free quantity but the existing scale, selected by the lab.

In units of  $E_0$  everything then becomes a pure  $\xi$ -power times  $\frac{1}{2}$ :

$$\frac{\omega}{E_0} = \frac{\xi}{2}, \quad \frac{m_{\text{window}}}{E_0} = \frac{\xi^2}{2}. \quad (2)$$

No free parameter remains. The  $\frac{1}{2}$  is the velocity square  $(\frac{1}{2}(m/p)^2)$ , Doc. 047) – kinematic, not a tuning factor; its *structural* origin is the AM sideband factor (see the signal-engineering reading below).

**What honestly remains.** The lab window (49–800 meV) is a factor  $\sim$ few wide: it selects the *scale*  $E_0$  sharply (excluding the alternatives by orders of magnitude) but does not pin the exact coefficient by itself. What remains is exactly *one* structural statement,

$$\omega = \xi E_0, \quad (3)$$

–  $E_0$  suppressed by exactly *one* fractal  $\xi$ -power ( $D_f = 3 - \xi$ ). This is no longer “free scale, self-referential” but a non-circular computation step:  $\omega$  as the  $\xi$ -suppressed eigenvalue of the connection-Laplacian in the time-extended Hilbert structure (Doc. 283 §8). It closes the point – or falsifies it. Computed: `schliesse_omega.py`.

## .9 The $\xi^1$ mechanism: lifting the zero mode by the dimension deficit

The remaining residual – why  $\omega = \xi E_0$  with exactly *one*  $\xi$ -power – has a concrete mechanism. The connection-Laplacian on  $T^4$  has an exact zero mode (the holonomy-flat constant). The fractal correction  $g = \eta(1 + \xi f)$  acts as a diagonal potential perturbation and lifts it; first-order perturbation theory gives

$$\delta\lambda_0 = \xi \langle 0 | f | 0 \rangle = \xi \cdot \bar{f}, \quad (4)$$

i.e. the power  $\xi^1$  *precisely* when the correction has a non-vanishing torus average  $\bar{f} \neq 0$ ; for  $\bar{f} = 0$  the first order vanishes and the mode lifts only at  $\xi^2$ .

(Note of 2 Oct. 2026 on notation: here  $f$  in  $g = \eta(1 + \xi f)$  is the fractal correction function of the metric, not a frequency and not the base winding number  $f = 1/\xi$ . Cf. Doc. 190, R147.)

$D_f = 3 - \xi$  is a dimension deficit – a systematic, non-volume-preserving change of the measure –, hence  $\bar{f} \neq 0$ . So the power is *one*. Numerically (`xi1_eigenwert.py`): the zero-mode lift scales with  $p = 0.999$  for a deficit ( $\bar{f} \neq 0$ ) versus  $p = 1.98$  for a volume-preserving ripple ( $\bar{f} = 0$ ) – so the power is not accidental but follows from the nature of  $D_f = 3 - \xi$ .

**Status.** The  $\xi^1$  power is thus *derived* (zero-mode lift by the dimension deficit), not asserted. What remains is only the  $O(1)$  coefficient – kinematically the  $\frac{1}{2}$  (Doc. 047) – and confirming that the physical kinetic state is this lifted zero mode. With that,  $\omega = \frac{\xi}{2} E_0$  is closed up to an  $O(1)$  factor – and that  $O(1)$  factor is identified structurally, in the signal-engineering reading below, as the sideband  $\frac{1}{2}$ .

## .10 The signal-engineering reading: AM sidebands and mode-locking

FFGFT grew out of communications engineering and the Fourier duality time $\leftrightarrow$ frequency (Doc. 001a);  $\hat{T} \cdot m = 1$  is that duality. In the dynamic sector the same language yields two concrete statements – one solid, one a candidate.

**The  $\frac{1}{2}$  is the AM sideband factor.** The fractal correction  $g = \eta(1 + \xi f)$  is formally an amplitude modulation of the carrier with modulation index  $\xi$  (Doc. 057: multiplicative = AM, additive = FM). Marcel’s kinetic prefactor  $A(t) = 1 + \varepsilon(t)$ ,  $|\varepsilon| \ll 1$ , is the same object ( $\xi \leftrightarrow \varepsilon$ ). For a carrier  $\cos(\omega_c t)$  one obtains necessarily

$$\cos(\omega_c t) [1 + \xi \cos(\omega_m t)] = \cos(\omega_c t) + \frac{\xi}{2} [\cos((\omega_c + \omega_m)t) + \cos((\omega_c - \omega_m)t)], \quad (5)$$

so the sidebands sit at amplitude  $\xi/2$ . The kinetic scale  $\omega = \frac{\xi}{2}E_0$  is therefore the *first AM sideband of the mass carrier*  $E_0$ : the  $\xi$  from the modulation index (= dimension deficit, previous section), the  $\frac{1}{2}$  from the sideband split ( $\cos \cdot \cos = \frac{1}{2}(\text{sum} + \text{difference})$ ). This unifies the  $\xi^1$  power (lifted zero mode = first sideband) and the  $O(1)$  coefficient into *one* mechanism: the  $\frac{1}{2}$  is not inserted kinematically but is structural – the velocity square ( $\frac{1}{2}(m/p)^2$ , Doc. 047) is its kinematic shadow, not its origin.

**$\theta = 2/9$  as a mode-locking ratio (candidate).** The phase layer of the fork (Doc. 283) reads: *dynamic operator-phase*  $R$  (HLV, skew-adjoint, driven by  $\dot{\phi}$ ) against *fixed scalar phase*  $\theta = 2/9$  (FFGFT). Locking a dynamic phase onto a fixed rational value is well-defined in signal engineering: injection locking, i.e. Arnold tongues. A driven nonlinear oscillator locks onto rational rotation numbers  $p/q$ ; the stable plateaus form a devil's staircase.  $\theta = 2/9$  would then be the lock plateau of a rational rotation number – which connects directly to FFGFT's *winding numbers*  $(n_\theta, n_\phi, n_3, n_4)$  (Doc. 210): a locked rational winding. Mode-locking is a quantization rule – rational values are the stable fixed points – exactly the kind of “holonomy/fixed-point/quantization rule” the open  $\theta$  point demands. The *open static*  $\theta$  input would thereby acquire a *dynamic* origin in the same sector.

**What stays a candidate.** The sideband point is solid: it reproduces  $\xi/2$  necessarily. The mode-locking reading is a *mechanism candidate*, not a derivation. There is a tension: small denominators ( $1/2, 1/3, 2/5$ ) have the widest tongues, while  $2/9$  (denominator 9) would be a narrower plateau; *why exactly*  $2/9$  locks is open. But the reframing is productive: “why this value?” becomes “which Arnold tongue, and why is it stable?” – a testable question about a driven oscillator, not about a bare number. An honest bound on the candidate: the rotation number  $2/9$  is *dimensionless* (cycles), the Koide offset  $\theta = 2/9$  is a *radian* angle –  $2\pi \cdot 2/9 = 4\pi/9 \neq 2/9$  –; reading the two as the same is an additional, open assumption, not part of the computed candidate (cf. Doc. 282).

**$Z_3$  protection: the denominator is explained, the numerator is not.** The FFGFT mass core is a  $Z_3$  circulant (Doc. 282); if the driving nonlinearity carries this symmetry (invariant under  $\theta \rightarrow \theta + 1/3$ , i.e. only harmonics  $k = 3, 6, 9, \dots$ ), the picture shifts. A structurally faithful  $Z_3$  drive widens the tongues with 3-power denominators *selectively*: at equal criticality  $1/3$  becomes 24× wider and the whole 9-family 11–12×, while the control tongue  $2/5$  (denominator 5, not a multiple of 3) stays *flat* ( $0.97\times$ ) – the signature of genuine symmetry protection, not an artefact (`z3_modeLock_test.py`). This answers “why a denominator-9 lock at all”: the  $Z_3$  structure of the core;  $2/9$  is no longer absurdly narrow but symmetry-protected robust. *The numerator stays open*: the symmetry protects the 9-family jointly ( $1/9 = 2/9 = 4/9$  identical), it does not single out  $2/9$ , and  $1/3$  stays wider. The question thus splits cleanly – “why  $2/9$ ” = “why denominator 9” ( $Z_3$ , settled) + “why numerator 2 in the family” (open, exactly Marcel's BD17A holonomy domain).

**The numerator needs a holonomy.** The 9-family is a  $Z_3$  orbit; a single 3-fold drive is therefore numerator-blind. The numerator is selected only by a *second* 3-structure with a definite relative phase  $\phi$  – a holonomy that lives in the embedding space ( $3'$  of the doubling  $6 = 3 \oplus 3'$ ,  $T^4 \rightarrow T^7$ , Doc. 285), not in the core. With two 3-fold harmonics ( $h=3, h=9$ ) and a relative phase  $\phi$  (not gauge-removable) the degeneracy breaks (`z3_holonomie_test.py`, bisection):  $1/9$  and  $4/9$  stay paired,  $2/9$  is the singlet that moves – for  $\phi \in (\pi, 2\pi)$   $2/9$  is the *widest* tongue (clearest margin at  $\phi = 3\pi/2$ ), crossover exactly at  $\phi = \pi$ . *Honestly*: the effect is small ( $\sim 0.5\%$ ),  $\phi$  is free (FFGFT does not fix it), and the minimal model splits only

2-against-1. But the structural statement stands: a holonomy *can* select  $2/9$ , and  $2/9$  is exactly the member it acts on. The open question is thereby precisely shifted – no longer “why  $2/9$ ” but “what locks  $\phi$  into the band where  $2/9$  wins” – and that is Marcel’s BD17A “locked holonomy”.

**One fractal correction, two orthogonal phases.** Holonomy and time modulation spring from the *same* fractal correction  $g = \eta(1 + \xi f)$  – the driving nonlinearity *is*  $f$  (Doc. 051, Doc. 270). One might fear that treating them as independent is inconsistent. The computation (`eine_fraktale_quelle.py`, 3-adic Weierstrass function) shows the opposite: they are two *orthogonal* functionals of  $f$ . The time phase  $A = 2\pi\xi\langle f \rangle$  reads the *mean*  $\langle f \rangle$  (the DC part, the dimension deficit,  $O(\xi)$ ); the holonomy  $\phi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$  reads the *phases of the 3-power harmonics* ( $O(1)$ ). The two are analytically decoupled –  $\langle f \rangle$  is independent of the harmonic phase (numerically to  $2 \cdot 10^{-16}$ ), since  $\cos(k\theta + \psi)$  has vanishing mean for every  $\psi$ . The worry is thereby resolved: neglecting  $A$  does not touch  $\phi$ , they read different Fourier parts of the *same*  $f$ . The real consistency lives elsewhere:  $A$  and the  $\xi^1$  mass lift *both* depend on  $\langle f \rangle$  – so the time phase is tied to the mass window, not free. And no free lock: a phase-coherent fractal ( $\psi = 0$ ) gives  $\phi = 0$  (degeneracy), whereas  $2/9$  requires  $f$ ’s 3-power harmonics to carry a relative phase in  $(\pi, 2\pi)$  (a ramp  $\delta = \pi/2$  per octave hits  $\phi = 3\pi/2$  exactly, the  $2/9$  maximum – but as a choice, not a derivation). Thus “what locks  $\phi$ ” is concretely named: not some arbitrary phase, but the relative phase of the 3-power harmonics of the fractal correction  $f$  – the FFGFT-internal target for BD17A.

**Demarcation: what is not a mechanism here.** Modulation index and lock ratio are different quantities in different sectors:  $\xi = 4/30000$  is the fixed AM index in the amplitude sector (it yields the sideband  $\frac{1}{2}$ ),  $2/9$  is a rotation number in the phase sector. The lock ratio is *not*  $\xi$ . Any “derivation” that sets  $\xi$  to an  $O(1)$  value (such as  $1/3$  or  $\tan(\pi/9) \approx 0.364$ ) in order to force  $2/9$  has left FFGFT –  $\xi$  is fixed and orders of magnitude smaller ( $\sim 10^{-4}$ ). Likewise the open question is the *stability* of the  $2/9$  plateau, not an algebra between  $\xi$  and  $2/9$ ; and the memory kernel the locking would rest on is the discrete-spectral, oscillating kernel of Doc. 282 ( $\sum_k g_k^2 \delta(\omega - \omega_k)$ , with backflow/revivals), not a dispersive all-pass.

## .11 Recursion as feedback: loop gain, regimes, discipline

FFGFT grew out of communications engineering (Doc. 001a), and its recursions are *feedback loops*. This places the results above in a frame – and fixes what about them is *free* and what is not.

**The loop gain is a  $\xi$  quantity – and fixed.** The magnitude of the loop gain decides the behaviour. In FFGFT it is geometrically given, not a dial:

- $|v| < 1$  (negative feedback): the loop contracts. The  $\xi$ -damping recursion  $r(k+1) = r(k)(1 - \xi)$  (Doc. 275) has gain  $1 - \xi$ , only  $\xi$  below one – hence the enormous, slow scale tower; it runs to its only fixed point 0 and merely *passes through*  $1/\varphi$  (no attractor).
- $|v| = 1$  (Barkhausen): marginal sustained oscillation.
- $|v| > 1$  (positive feedback): the loop runs away. The golden  $\varphi$  branch ( $|\varphi| > 1$ ) unrolls, the  $-1/\varphi$  branch ( $|-1/\varphi| < 1$ ) is the overdamped empty window (Doc. 285).

Decompactification is exactly this transition: *one* circle tips from the compact (contracting) into the runaway regime – the mass circle  $\tilde{T} \cdot m = 1$  unrolls to non-compact time (Doc. 270), and that tipping *is* the measurement.

**The phase selects itself.** A feedback loop imposes no phase: the Barkhausen phase condition closes at the frequency the loop *finds*; in injection locking the lock phase slides within the lock range (Adler equation). Hence  $\theta = 2/9$  is to be read as a *self-selected lock ratio* (rotation number), not as a tightly imposed phase. A "derivation" that prescribes a fixed phase  $2/9$  and solves backwards for  $\xi$  mistakes this self-adjustment (cf. the demarcation above).

**Two tiers.** Linear and weak ( $\xi$  small): first order = the AM sideband  $\frac{1}{2}$  (above, solid); a purely linear loop does not lock. Nonlinear: mode-locking onto rational rotation numbers = the  $\theta = 2/9$  candidate (above); the lock-range width scales with the feedback strength  $\xi$ , and  $\xi \sim 10^{-4}$  with denominator 9 makes the  $2/9$  tongue extremely narrow – that is the open stability problem, not a missing piece of algebra.

**Negative feedback as equalization.** The fractal correction  $g = \eta(1 + \xi f)$  (with  $K_{\text{frak}}/\text{RG}$ , Doc. 133/051) acts as broadband negative feedback that linearizes the bare carrier and sets  $D_f = 3 - \xi$  – weak, with no sharp phase alignment required.

**Discipline.** This fixes how the tools are to be used *correctly*: (i) the loop gain ( $\xi$ , the eigenvalues  $\varphi/-1/\varphi$ ) is fixed, never a dial – forcing a result by setting  $\xi$  is the cardinal error; (ii) the open questions are always *regime and lock range* (contracting/marginal/runaway; tongue width), never an algebra on the gain; (iii) the phase is self-selected, quantized values are lock ratios; (iv) the regime of the feedback carries the physics – negative feedback = compact stable scales + equalization, positive feedback = decompactification/measurement.

## .12 Relation to the static sector

The static sector is closed and computed: the  $Z_3$  mass operator ( $r = \sqrt{2}$ ,  $\theta = 2/9$ ) plus the mass circle yields Koide  $2/3$ , the lepton ratios and – as a pre-registered, matched-null-tested declaration – the BD12-XCORE interface (Doc. 285). The dynamic sector is the *complementary, open* side: the same parameter  $\xi$ , but as a *fraction* rather than an *eigenvalue*. A derivation of  $m_{\text{window}} = \xi \cdot \omega$  would move the window-filling dynamics (Doc. 263, holographic scale projection) from description to computation, and the neutrino-like  $3'$  candidate (Doc. 285) from conditional to derived.

**Retrospect: two things both called " $2/9$ ".** It is worth fixing the thread, for  $\theta = 2/9$  appears in two roles. (i)  $2/9$  as a *number/input*: the  $Z_3$  circulant of the core uses it and reproduces the leptons (Koide), empirically closed – it was always there (Doc. 282; Doc. 285, the "protected-invariant" row  $r = \sqrt{2}$ ,  $Q = 2/3$ ,  $\theta = 2/9$ ). (ii)  $2/9$  as a *question* ("why this value", specifically the numerator): this becomes relevant only with the dimensional extension. The reason is structural, not a lack of computation: the bare  $Z_3$  core is *numerator-blind* –  $1/9, 2/9, 4/9$  are a  $Z_3$  orbit, and a single compact 3-structure cannot in principle answer "why  $2/9$  rather than  $1/9$ ". The numerator is seen only by a second 3-structure with a relative phase = a holonomy = exactly what the extension provides ( $3 \oplus 3', T^4 \hookrightarrow T^7$ ). The "why- $2/9$ " question thus has no home in the core; it is by nature an extension question.

**Where this already stands in the corpus.** Three documents already carry the thread. **Doc. 285** (dimension bridge) is the extension itself:  $\text{HLV} \supset \text{FFGFT}$ ,  $T^4 \hookrightarrow T^7$  ( $7 = 4 + 3$ , the three perp dimensions), the doubling  $6 = 3 \oplus 3'$  ("literally"), the fork (FFGFT 3-fold

$\sqrt{2}/Q = 2/3/\theta = 2/9$  against HLV 5-fold  $2/\sqrt{5}/\varphi$ ;  $\theta = 2/9$  sits in the invariant row. **Doc. 283** (bridge and fork, §8) introduces  $\theta = 2/9$  explicitly as a *complex holonomy*, the “operational consequence of compactness”, with the phase layer “dynamic operator-phase  $R$  (HLV) against fixed scalar phase  $\theta = 2/9$  (FFGFT)” – there  $2/9$  is already a holonomy/extension object, not a bare core number. **Doc. 282** is the core that *uses*  $2/9$  (leptons), with the honest line “translation is not justification” – the notation *represents*  $2/9$  but does not *ground* it.

**The fixed point and the conclusion.** Why the static sector is closed *at all*: the lepton masses sit at a *self-consistent fixed point* – the mass operator  $\Phi_* = \sum_i m_i P_i$  is the non-trivial fixed point of the recursion  $\mathcal{R}(\Phi_*) = \Phi_*$  (Doc. 203, lepton exponents  $p_e = 3/2$ ,  $p_\mu = 1$ ,  $p_\tau = 2/3$ ); *that* is why it is exact, not a fitted value. From the lepton measurements one *reads off* this fixed point (that is the closure); fixed and open separate cleanly:  $Q = 2/3$  is the operator condition  $r = \sqrt{2}$  (algebraically fixed, Doc. 282),  $\theta = 2/9$  is the holonomy (the numerator, open). Hence the conclusion: all the recent work above (feedback, mode-locking, holonomy test) is *no new arena* but the mechanistic, signal-engineering version of what  $282 \rightarrow 283 \rightarrow 285$  already say structurally –  $2/9$  is a holonomy, and a holonomy lives in the extension.

**Magnitude and phase: why the split is inevitable.** In signal-engineering language the whole finding collapses to a single, well-known theorem. The mass operator *is* a  $Z_3$  circulant, and a circulant is diagonalized by the DFT – its eigenvalues are the DFT of its first row. The three lepton masses are thus literally a *three-point spectrum*:  $\sqrt{2}$  is the *magnitude* of the spectral coefficients,  $\theta = 2/9$  their *phase*. The radial/angular conclusion (Doc. 282) is therefore nothing but *magnitude spectrum versus phase spectrum*: everything determined sits in the magnitude ( $\xi$  as scale, the hierarchy as envelope,  $Q = 2/3 = \sqrt{2}$  as amplitude ratio,  $\langle f \rangle$  as DC term), everything open in the phase ( $\theta$ , the holonomy  $\varphi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$ , the harmonic phases). The deep reason is the theorem itself: *a magnitude spectrum does not fix the phase spectrum* – same power, arbitrary shape. *That* is precisely why  $\theta$  resists every FFGFT tool: the recursion  $\mathcal{R}$  is purely real, hence pure gain, and real gain does not touch the phase. The orthogonality computed above *is* “magnitude  $\perp$  phase”. From this the FFGFT $\leftrightarrow$ HLV division of labour even becomes a circuit diagram: FFGFT is the *real amplifier stage* (magnitude, hierarchy), HLV’s skew-adjoint  $R$  is the *all-pass/phase stage* ( $e^{i\varphi}$ , magnitude-preserving, rotating only the phase). A skew-adjoint operation *is* an all-pass. The missing angular principle is the phase stage – and the HLV side carries it, not FFGFT. (Limit of the same lens: mode-locking yields a *rotation number* (dimensionless),  $\theta$  a *radian*;  $2\pi \cdot 2/9 \neq 2/9$  – the lens identifies the *type* correctly, a locked phase, but the literal identification does not close.)

## .13 Reproducibility

kinetik\_xi\_anteil.py (numpy-only): narrowing of the  $\xi$ -power against the bounds, the duality framing ( $\tilde{T} = 1/\max(m, \omega)$ ), and the finding  $p = 2$  with the explicitly open reference/-coefficient/spectrum gap. The script claims no derivation; it narrows. schliesse\_omega.py: the lab selection of the reference  $E_0$  (table) and the reduction  $\omega/E_0 = \xi/2$ ,  $m_{\text{window}}/E_0 = \xi^2/2$ , with the  $\xi^1$  remainder as the marked closing point. xi1\_eigenwert.py: the zero-mode lift as  $\xi^1$  from the dimension deficit ( $p = 0.999$  vs  $p = 1.98$ ). am\_seitenband\_modeLock.py: confirms the sideband ratio  $\xi/2$  necessarily (deviation 0%) and measures the Arnold-tongue widths on the sine circle map –  $2/9$  locks as a plateau but is  $\sim 12\times$  narrower than  $1/3$ : candidate, not derivation. z3\_modeLock\_test.py: a  $Z_3$ -symmetric drive (harmonics = multiples of 3) widens the 3-power-denominator tongues selectively ( $1/3$  by  $24\times$ , the 9-family by  $11\text{--}12\times$ ), while the control  $2/5$  stays flat ( $0.97\times$ ) – the denominator 9 is  $Z_3$ -protected (Doc. 282), the numerator 2

stays open. `z3_holonomie_test.py`: two 3-fold harmonics ( $h=3, h=9$ ) with a relative phase  $\phi$ ; measured precisely (bisection),  $\phi$  breaks the 9-degeneracy –  $1/9, 4/9$  stay paired,  $2/9$  moves and is the widest tongue for  $\phi \in (\pi, 2\pi)$  ( $\sim 0.5\%$ , maximum at  $\phi = 3\pi/2$ );  $\phi$  free  $\Rightarrow$  shifts the open question to the  $\phi$ -lock (BD17A). `eine_fraktale_quelle.py`: shows computationally (3-adic Weierstrass  $f$ ) that the time phase and the holonomy are *orthogonal* functionals of the same fractal correction –  $A = 2\pi\xi\langle f \rangle$  reads the mean ( $O(\xi)$ ),  $\phi = \arg \hat{f}_9 - 3 \arg \hat{f}_3$  the harmonic phases ( $O(1)$ ),  $\langle f \rangle$  phase-independent to  $2 \cdot 10^{-16}$ ; the time phase is tied to the  $\xi^1$  mass lift (both  $\langle f \rangle$ ), the holonomy phase stays an independent datum (coherent  $\Rightarrow \phi = 0$ ;  $\delta = \pi/2$  ramp  $\Rightarrow \phi = 3\pi/2$ ).